

The equation and its root

An equality that is true only for certain values — and what it means to check one.

LESSON 122

1 × 40 MIN

MATEMATIKA 7, §54

S8 2.5

LESSON CLOCK

10'

10'

12'

6'

2'

Three kinds of equality	10 min
What a root is	10 min
How many roots	12 min
Equivalent equations	6 min
Homework	2 min

LEARNING OBJECTIVES

- Distinguish an equation from an identity and from an expression.
- Define a root and verify one by substitution.
- Say how many roots an equation has, including none and infinitely many.
- Recognise equivalent equations.

TEXTBOOK

National	pp. 358–362
Cambridge	Stage 8, pp. 31–36
Workbook	Exercise 2.5

01

Explanation

Three kinds of equality

WRITTEN	NAME	TRUE FOR
$2x + 3$	an expression	nothing — it makes no claim
$2x + 3 = 11$	an equation	one value, $x = 4$
$(a + b)^2 = a^2 + 2ab + b^2$	an identity	every value

AN EQUATION IS A QUESTION

It asks: for which values of the letter is this equality true? The whole of the rest of the quarter is about answering that question efficiently.

What a root is

A **root** is a value of the unknown that makes the equation a true statement. Substituting it is the only test.

EQUATION	CANDIDATE	LEFT SIDE	RIGHT SIDE	A ROOT?
$2x + 3 = 11$	$x = 4$	11	11	yes
$2x + 3 = 11$	$x = 3$	9	11	no
$x^2 = 9$	$x = -3$	9	9	yes
$x^2 = 9$	$x = 3$	9	9	yes — there are two

CHECKING MEANS SUBSTITUTING INTO THE ORIGINAL EQUATION

Substituting into a line halfway through the working repeats any error made before it. Go back to the equation as it was set.

How many roots

EQUATION	ROOTS	HOW MANY
$2x + 3 = 11$	$x = 4$	one
$x^2 = 9$	$x = \pm 3$	two
$x + 1 = x + 2$	none	none — it says $1 = 2$
$2(x + 1) = 2x + 2$	every number	infinitely many — an identity
$x^2 = -4$	none among the numbers we know	none

“NO ROOTS” IS AN ANSWER

Reaching $0 = 1$ does not mean the working failed; it means the equation is never true. Reaching $0 = 0$ means it is always true. Both are complete answers.

Equivalent equations

Two equations are **equivalent** when they have exactly the same roots. Solving an equation is replacing it by simpler equivalent ones until the answer is visible.

OPERATION	KEEPS THE ROOTS?
add the same number to both sides	yes
multiply both sides by a non-zero number	yes
multiply both sides by 0	no — everything becomes a root
square both sides	not always — new roots can appear

MULTIPLYING BY AN EXPRESSION THAT COULD BE ZERO IS NOT SAFE

Multiplying $x = 1$ by x gives $x^2 = x$, which also has the root 0 . The new equation is not equivalent, and that is why every answer is checked.

02

Worked examples

EXAMPLE 1 Is $x = 4$ a root of $2x + 3 = 11$?

1

Substitute: $2 \cdot 4 + 3$.

2

$= 11$

3

The right side is 11 .

4

The equality holds, so yes.

ANSWER

Yes

EXAMPLE 2 How many roots has $x + 1 = x + 2$?

- ① Subtract x from both sides.

- ② $1 = 2$

- ③ This is false for every value of x .

- ④ The equation has no roots.

ANSWER

None

EXAMPLE 3 How many roots has $2(x + 1) = 2x + 2$?

- ① Expand the left side: $2x + 2$.

- ② $2x + 2 = 2x + 2$

- ③ True for every value of x .

- ④ It is an identity — infinitely many roots.

ANSWER

Every number

03 Interactive model

Give three equations with one root, no roots and infinitely many, unlabelled, and ask the class to sort them; the categories are learnt in one go.

Root, no root, or every number?

Is $x = 4$ a root of $2x + 3 = 11$?

Is $x = 3$ a root of $2x + 3 = 11$?

How many roots has $x^2 = 9$?

How many roots has $x + 1 = x + 2$?

0

1

2

infinitely many

How many roots has $2(x + 1) = 2x + 2$?

0

1

2

infinitely many

$2x + 3$ on its own is:

an equation

an expression

an identity

a root

Reaching $0 = 0$ means:

no roots

one root

every number is a root

an error

Squaring both sides:

always keeps the roots

can add new ones

removes roots

is never allowed

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Equation	Tenglama	Уравнение
Root	Ildiz	Корень
Solution	Yechim	Решение
To solve	Yechmoq	Решить
Unknown	Noma'lum	Неизвестное
Identity	Ayniyat	Тождество
To check	Tekshirmoq	Проверить
Equivalent equations	Teng kuchli tenglamalar	Равносильные уравнения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A root is a value that makes the equation:

A root is checked by:

$x^2 = 9$ has:

An identity is true for:

one value

two values

every value

no value

Reaching $0 = 1$ means:

no roots

every root

one root

an error was made

Equivalent equations have:

the same form

the same roots

the same degree

the same letters

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Is $x = 4$ a root of $2x + 3 = 11$?

ANSWER Yes

2. Is $x = 3$ a root of $2x + 3 = 11$?

ANSWER No

3. Is $x = -3$ a root of $x^2 = 9$?

ANSWER Yes

4. The roots of $x^2 = 9$

ANSWER $x = \pm 3$

5. The root of $x + 5 = 12$

ANSWER $x = 7$

6. The root of $3x = 12$

ANSWER $x = 4$

7. $2x + 3$ is an equation or an expression?

ANSWER An expression

Medium · the standard question

7 PROBLEMS

1. How many roots has $x + 1 = x + 2$?

ANSWER None

2. How many roots has $2(x + 1) = 2x + 2$?

ANSWER Infinitely many

3. Is $(a + b)^2 = a^2 + 2ab + b^2$ an equation or an identity?

ANSWER An identity

4. Is $x = 1$ a root of $x^2 = x$?

ANSWER Yes

5. Is $x = 0$ a root of $x^2 = x$?

ANSWER Yes

6. Is $x = 0$ a root of $x = 1$?

ANSWER No

7. What does reaching $0 = 5$ tell you?

ANSWER The equation has no roots

Hard · stretch and reason

7 PROBLEMS

1. Are $x = 1$ and $x^2 = x$ equivalent?

ANSWER No — the second also has the root 0

2. Which operations preserve the roots?

ANSWER Adding the same number, and multiplying by a non-zero number

3. Why is multiplying both sides by x unsafe?

ANSWER x may be zero, and a new root appears

4. How many roots has $x^2 = -4$ among the numbers of Grade 7?

ANSWER None

5. Give an equation with exactly one root

ANSWER e.g. $2x = 6$

6. Give an equation with no roots

ANSWER e.g. $x = x + 1$

7. Give an equation true for every number

ANSWER e.g. $3(x - 1) = 3x - 3$

07 Homework

Homework — 5 tasks

Check every answer by substituting into the equation as it was originally written.

1. Check whether $x = 5$ is a root of $3x - 4 = 11$.
2. Find all the roots of $x^2 = 16$.
3. How many roots has $x + 3 = x + 7$? Explain.
4. How many roots has $5(x - 2) = 5x - 10$? Explain.
5. Write one equation with one root, one with none, and one true for every number.